Linux Commands Assignments - II

1. List the directory contents date wise sorted.

```
rahulastic@IDEAPAD:~$ ls -lt
total 44
-rw-r--r-- 1 rahulastic rahulastic   21 Oct 28 10:38 numbers.txt
-rw-r--r-- 1 rahulastic rahulastic   39 Oct 28 10:35 fruits.txt
-rw-r--r-- 1 rahulastic rahulastic    0 Oct 28 10:31 a.txt
-rw-r--r-- 1 rahulastic rahulastic    0 Oct 28 10:31 b.txt
-rw-r--r-- 1 rahulastic rahulastic    0 Oct 28 10:31 c.txt
-rw-r--r-- 1 rahulastic rahulastic   58 Oct 28 10:30 myfile.txt
drwxr-xr-x 2 rahulastic rahulastic 4096 Oct 28 10:20 new_directory
drwxr-xr-x 2 rahulastic rahulastic 4096 Oct 28 10:05 backup_dir
-rw-r--r-- 1 rahulastic rahulastic  120 Oct 15 09:40 test.txt
-rw-r--r-- 1 rahulastic rahulastic   12 Oct 12 18:22 hello.txt
-rw-r--r-- 1 rahulastic rahulastic   30 Oct 12 18:10 filename.txt
-rw-r--r-- 1 rahulastic rahulastic   85 Oct 10 11:05 demo01.txt
-rw-r--r-- 1 rahulastic rahulastic 2150 Oct  9 16:30 vim
-rw-r--r-- 1 rahulastic rahulastic 1180 Oct  9 16:12 editor
```

2. List the directory contents size wise sorted.

```
rahulastic@IDEAPAD:~$ ls -lS
total 44
drwxr-xr-x 2 rahulastic rahulastic 4096 Oct 28 10:05 backup_dir
drwxr-xr-x 2 rahulastic rahulastic 4096 Oct 28 10:20 new_directory
-rw-r--r-- 1 rahulastic rahulastic 2150 Oct  9 16:30 vim
-rw-r--r-- 1 rahulastic rahulastic 1180 Oct  9 16:12 editor
-rw-r--r-- 1 rahulastic rahulastic  120 Oct 15 09:40 test.txt
-rw-r--r-- 1 rahulastic rahulastic   85 Oct 10 11:05 demo01.txt
-rw-r--r-- 1 rahulastic rahulastic   58 Oct 28 10:30 myfile.txt
-rw-r--r-- 1 rahulastic rahulastic   39 Oct 28 10:35 fruits.txt
-rw-r--r-- 1 rahulastic rahulastic   30 Oct 12 18:10 filename.txt
-rw-r--r-- 1 rahulastic rahulastic   21 Oct 28 10:38 numbers.txt
-rw-r--r-- 1 rahulastic rahulastic   12 Oct 12 18:22 hello.txt
-rw-r--r-- 1 rahulastic rahulastic    0 Oct 28 10:31 a.txt
-rw-r--r-- 1 rahulastic rahulastic    0 Oct 28 10:31 b.txt
-rw-r--r-- 1 rahulastic rahulastic    0 Oct 28 10:31 c.txt
```

3. List directory contents along with their inode no.

```
rahulastic@IDEAPAD:~$ ls -i
524710 a.txt      524301 editor       524322 new_directory
524711 b.txt      524305 filename.txt  524719 numbers.txt
524330 backup_dir  524718 fruits.txt    524311 test.txt
524712 c.txt      524307 hello.txt     524303 vim
524298 demo01.txt  524702 myfile.txt
```

4. List the contents of the subdirectory.

```
rahulastic@IDEAPAD:~$ ls new_directory
rahulastic@IDEAPAD:~$ ls -a new_directory
.  ..
rahulastic@IDEAPAD:~$ ls -l new_directory
total 0
```

5. Create a file, write your surname and name in it and save.

```
rahulastic@IDEAPAD:~$ cat > details.txt
<Your surname> Rahul
rahulastic@IDEAPAD:~$ cat details.txt
<Your surname> Rahul
```

i. Open the same file and add your address in it and save.

```
rahulastic@IDEAPAD:~$ cat >> details.txt
<Your address>
```

ii. Reopen the same file and check your name and address in it.

```
rahulastic@IDEAPAD:~$ cat details.txt
<Your surname> Rahul
<Your address>
```

6. How to change the time stamp of a file.

```
rahulastic@IDEAPAD:~$ ls -l details.txt
-rw-r--r-- 1 rahulastic rahulastic 36 Oct 28 11:02 details.txt
rahulastic@IDEAPAD:~$ touch details.txt
rahulastic@IDEAPAD:~$ ls -l details.txt
-rw-r--r-- 1 rahulastic rahulastic 36 Oct 28 11:10 details.txt
rahulastic@IDEAPAD:~$ touch -t 202610011030 details.txt
rahulastic@IDEAPAD:~$ ls -l details.txt
-rw-r--r-- 1 rahulastic rahulastic 36 Oct  1 10:30 details.txt
rahulastic@IDEAPAD:~$ touch -d "2026-09-15 09:00" details.txt
rahulastic@IDEAPAD:~$ ls -l details.txt
-rw-r--r-- 1 rahulastic rahulastic 36 Sep 15 09:00 details.txt
```

7. Create the directory structure using the given commands.

```
rahulastic@IDEAPAD:~$ mkdir one
rahulastic@IDEAPAD:~$ cd one
rahulastic@IDEAPAD:~/one$ touch 1.txt 11.txt 111.txt
rahulastic@IDEAPAD:~/one$ mkdir two
rahulastic@IDEAPAD:~/one$ cd two
rahulastic@IDEAPAD:~/one/two$ touch 2.txt 22.txt 222.txt
rahulastic@IDEAPAD:~/one/two$ mkdir three
rahulastic@IDEAPAD:~/one/two$ cd three
rahulastic@IDEAPAD:~/one/two/three$ touch 3.txt 33.txt 333.txt
rahulastic@IDEAPAD:~/one/two/three$ mkdir four
rahulastic@IDEAPAD:~/one/two/three$ cd four
rahulastic@IDEAPAD:~/one/two/three/four$ touch 4.txt 44.txt 444.txt
rahulastic@IDEAPAD:~/one/two/three/four$ mkdir five
rahulastic@IDEAPAD:~/one/two/three/four$ cd five
rahulastic@IDEAPAD:~/one/two/three/four/five$ touch 5.txt 55.txt 555.txt
rahulastic@IDEAPAD:~/one/two/three/four/five$ cd ~
rahulastic@IDEAPAD:~$ pwd
/home/rahulastic
```

8. Considering the above directory structure, do the following operations.

i. List the contents of directory "five" from the current directory (home).

```
rahulastic@IDEAPAD:~$ ls one/two/three/four/five
5.txt  55.txt  555.txt
```

ii. Write your name in file "44.txt" of directory "four".

```
rahulastic@IDEAPAD:~$ echo Rahul > one/two/three/four/44.txt
rahulastic@IDEAPAD:~$ cat one/two/three/four/44.txt
Rahul
```

iii. Remove the file "555.txt" from directory "five".

```
rahulastic@IDEAPAD:~$ rm one/two/three/four/five/555.txt
rahulastic@IDEAPAD:~$ ls one/two/three/four/five
5.txt  55.txt
```

iv. Change directory to five.

```
rahulastic@IDEAPAD:~$ cd one/two/three/four/five/
rahulastic@IDEAPAD:~/one/two/three/four/five$ pwd
/home/rahulastic/one/two/three/four/five
```

v. Write your course name in "3.txt" of directory "three" and read it.

```
rahulastic@IDEAPAD:~/one/two/three/four/five$ echo "Master of Computer
Management" > ../../3.txt
rahulastic@IDEAPAD:~/one/two/three/four/five$ cat ../../3.txt
Master of Computer Management
```

vi. List the contents of directory "two" from the current directory.

```
rahulastic@IDEAPAD:~/one/two/three/four/five$ ls ../../..
2.txt  22.txt  222.txt  three
```

vii. Remove file "222.txt" which belongs to directory "two".

```
rahulastic@IDEAPAD:~/one/two/three/four/five$ rm ../../../222.txt
rahulastic@IDEAPAD:~/one/two/three/four/five$ ls ../../..
2.txt  22.txt  three
```

viii. Change the directory to "one".

```
rahulastic@IDEAPAD:~/one/two/three/four/five$ cd ../../../..
rahulastic@IDEAPAD:~/one$ pwd
/home/rahulastic/one
```

ix. Remove all files only from directory "five".

```
rahulastic@IDEAPAD:~/one$ rm two/three/four/five/*
rahulastic@IDEAPAD:~/one$ ls two/three/four/five
```

x. Remove directory "five".

```
rahulastic@IDEAPAD:~/one$ rmdir two/three/four/five
rahulastic@IDEAPAD:~/one$ ls two/three/four
4.txt  44.txt  444.txt
```

xi. Remove the whole directory "four".

```
rahulastic@IDEAPAD:~/one$ rm -R two/three/four
rahulastic@IDEAPAD:~/one$ ls two/three
3.txt  33.txt  333.txt
```

xii. Change to your home directory.

```
rahulastic@IDEAPAD:~/one$ cd ~
rahulastic@IDEAPAD:~$ pwd
/home/rahulastic
```

9. Change the time stamp of file "3.txt" which resides in directory "three".

```
rahulastic@IDEAPAD:~$ ls -l one/two/three/3.txt
-rw-r--r-- 1 rahulastic rahulastic 30 Oct 28 11:25 one/two/three/3.txt
rahulastic@IDEAPAD:~$ touch one/two/three/3.txt
rahulastic@IDEAPAD:~$ ls -l one/two/three/3.txt
-rw-r--r-- 1 rahulastic rahulastic 30 Oct 28 11:40 one/two/three/3.txt
rahulastic@IDEAPAD:~$ touch -t 202609150900 one/two/three/3.txt
rahulastic@IDEAPAD:~$ ls -l one/two/three/3.txt
-rw-r--r-- 1 rahulastic rahulastic 30 Sep 15 09:00 one/two/three/3.txt
```